

Predictive Modeling of Financial Indicators in Singapore's Banking Sector using Deep Learning Architectures (LSTM)

Team Banklytics

Date: 10/12/2025

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Abstract

This report dives deeply into prediction of market trends of Singapore's major banking stocks (DBS, OCBC and UOB). A stacked Long Short-Term Memory (LSTM) neural network deep learning model with financial time series indicators and sentimental analytics was used as the primary framework to predict trends. Data from Yahoo Finance is combined with technical indicators and sentiment scores to develop a high-dimensional feature space. Log returns were used as modeling targets to improve stability and better understand market trends. This report contains details on the model architecture and design. The performance of the model is evaluated using a walk-forward expanding-window validation which retrains and tests the model. The model's performance is evaluated based on its ability to predict movements accurately. This report references the findings by Wang et al. (2024). The findings explain the benefits of using the LSTM model and how the walk-forward method provides a real-world assessment of financial time series data. The strengths and limitations of using LSTM based forecasting model are highlighted.

1. Introduction

Singapore's major banks are DBS, OCBC and UOB which are the backbone of the economy. They contributed significantly to the countries economic stability. Understanding the movements of their stock prices helps analyst predict market trends, macroeconomic factors, and global events. However, prediction of stock prices can be troublesome as past trends may not have any effect on current market behaviour.

The project aims to predict the stock price of major banks using deep learning-based models. With the help of technical indicators and sentiment scores, the model can predict short-term log returns. Python and R code are used in this report.

1.1 Project objectives

1. Identify an optimal combination of technical indicators. Technical indicators like trend indicators (MACD, moving average), momentum oscillators (RSI, Stochastic), volatility indicators (Bollinger bands, ATR), and volume-based measures (OBV, CMF).
2. Integrating sentimental scores. Converting sentiment news into positive, negative, or neutral scores to then add to the respective stocks to help with prediction.
3. Use log returns for the modeling target. Logarithmic return stabilizes the variance and simplifies the mathematical handling, which is better for forecasting compared to the raw stock price. This methodological choice is supported by Ciganović and D'Amario

(2024), who forecast cryptocurrency log returns using LASSO-VAR and sentiment signals

4. Develop an LSTM model for prediction. LSTM model is suitable for time series data as it has a memory gate.
5. The accuracy of the model is calculated using walk-forward expanding-window technique. This approach was referenced in Wang et al. (2024)
6. Directional accuracy as an evaluation metric. The accuracy is based on movement of the predicted and actual value.

2. Methodology

This section describes the data collection, feature engineering, model architecture, mathematical foundations, training techniques, and walk-forward testing design.

2.1 Data collection

Open, High, Low, Close, and Volume (OHLCV) data were collected from two main sources which are yFinance. Sentimental analysis data was collected from the GDELT API. The stock codes to get them are U11.SI for UOB, D05.SI for DBS, O39.SI and OCBC. The period = "max" gives the historical data for that stock.

```
yf.Ticker(symbol).history(period='max')
```

Sentiment data was extracted for the year 2024 and 2025. To classify the data into positive, negative, and neutral labeled based on the respective news. We used Vader based on the respective news. Some days had multiple labels and hence needed to be merged.

```
sentimental_data.groupby(["Date", "bank"])["count"].sum().reset_index()
```

To do so, group by and sum command was performed to have 1 to 1 relation between date and sentiment score.

2.2 Target variable: Logarithmic return

The model predicts the log return value for the next day.

```
df['Log_Ret'] = np.log(df['Close'] / df['Close'].shift(1))
```

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right)$$

The advantage of using this approach is the model's learning stability by behaving closer to stationery. Ciganović and D'Amario (2024) treat log returns as the primary predictive target, with final prices reconstructed using the exponential property of log returns.

2.3 Feature engineering

To understand which indicators best help predict market trends, all were included initially. Linear regression using R was performed on the dataset with all technical indicators to see how much significance each indicator shows with the log return price variable.

2.4 MinMax scaling

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

MinMax scaling was performed to normalize the data as neural networks are sensitive to varying magnitudes. An example is indicators like OBV values could be in the millions. Log returns values are usually between -0.02 to +0.02. MinMax scales the values of all features between a fixed range of [0-1].

2.5 LSTM Model Architecture

Figure 5 shows the model architecture.

1. The model takes 60-day rolling sequences as input with log returns, technical indicators, and sentiment scores.
2. The first LSTM layer has 50 units and returns the full sequence, allowing it to learn short- and medium-term temporal patterns.
3. A dropout layer is applied to tackle overfitting and improve generalization
4. The second LSTM layer also has 50 units but returns only the final state, enabling the model to capture long-term dependencies over the 60-day window
5. Another dropout layer further regularizes the network
6. Finally, a dense layer with a single neuron output the next-day log return prediction.

Two LSTM layers were used to capture both short-term and long-term temporal dependencies within the 60-day window, while dropout was applied after each layer to reduce overfitting and improve generalization in a noisy financial environment.

This architecture allows the model to learn both local and long-range behaviors in market data, which makes it suitable for financial time-series forecasting

2.6 Sequence construction

60 days chosen to capture multi-week technical indicator behavior, momentum trends, and market regimes while maintaining stable LSTM training.

A sliding window of 60 days is used to create sequence

$$X[i] = \text{features}[i : i+60] \quad y[i] = \text{target}[i+60]$$

This gives input tensors of shape: (60, number of features)

2.7 Walk-forward Expanding-Window validation

The model is evaluated using walk-forward expanding window validation, which is like real-world forecasting. Below are the steps:

1. Train the model using data from 2024-01-01 to the current validation date.
2. First, we choose a start date for predictions. Which is 2025-01-01
3. Next, the model generates a one-day-ahead forecast by predicting the next day's log return and converting it to a price.
4. We evaluate performance using directional accuracy by checking whether the predicted direction matches the actual movement.
5. After each prediction, the window shifts forward by one week, and the process repeats
6. This continues until the end of the test period, and we compute the overall average accuracy.

“Walk-forward machine learning offers a realistic and unbiased evaluation for time-series trading strategies by ensuring models are trained strictly on past data while being tested on unseen forward periods.” Wang et al. (2024)

2.8 Price reconstruction from log returns

Predicted log returns are converted back to prices: $P_{t+1} = P_t \cdot e^{\widehat{r_{t+1}}}$

This makes forecast outputs interpretable in real price terms.

2.9 Directional accuracy for evaluating model

Fig 4 shows how the flow of directional accuracy is evaluated.

Directional accuracy measures whether: $\text{sign}(P_{t+1} - P_{start}) = \text{sign}(\widehat{P_{t+1}} - P_{start})$

Expanding window training (walk-forward validation). Predict 1 day ahead.

$$\text{Accuracy} = \frac{\text{Correct up/down predictions}}{\text{Total predictions}}$$

For each day in the test window, it checks: Did the actual price end above or below the start price? Did the predicted price end above or below the start price? If both sides are on

the same side of start_price (both up or both down), that day is counted as correct trend direction.

3. Key findings and results.

This section explains what was achieved by the model and how certain selections played a part in the model's overall performance.

3.1 Summary of findings

The LSTM-based forecasting framework successfully integrates log returns, technical indicators, sentiment analytics, and a walk-forward evaluation procedure. Model performance is evaluated using a directional accuracy metric that compares the predicted and actual price movements to the last known closing price at the start of each test window. For each day in the evaluation period, the prediction is counted as correct if it correctly predicts whether the price moves above or below this reference level. The resulting average accuracies were **DBS: 0.432, OCBC: 0.470 and UOB: 0.500**

These values indicate that the model achieves directional accuracy close to 50% for all three stocks, which is above random guessing. The model seems to predict UOB better compared to other bank stocks, which indicates predictable behavior. According to The Straits Times (2025), UOB's third-quarter 2025 net profit fell 72% year-on-year to S\$443 million, largely due to a deliberate increase in credit provisions. Hence, this explains why the model achieved the highest directional accuracy among the three banks.

3.2 Feature combinations

Fig 1,2, 3 and table 3 are the outputs of **linear regression**. It shows significance with log return as the dependable variable. From the significance of each feature, ROC, TEMA, BB_High, MOM, MACD_Hist, EMA, SMA for all three banks. tenkan_sen and OBV features for OCBC and DBS were significant. **LASSO**: There were only two indicators that didn't shrink to zero (Refer to table 2). The two indicators that were selected were the Momentum indicator (MOM) and the Rate of Change indicator (ROC).

The sentiment score didn't show significance and was removed from the feature set. However, during the LSTM testing, the model's overall performance declined after the removal of sentiment score. This suggests that although sentiment score showed no significance with log returns, it might show nonlinear and time dependence that the LSTM model understands. So, the final feature combination selected for training the model is MOM, ROC and Sentiment Score.

3.3 Strengths of the approach

Combines multiple sources of market information, uses log returns because it is more stable, LSTM captures nonlinear and temporal dependencies, walk-forward expanding-window testing removes look-ahead bias, directional accuracy matches real trading goals and sentiment helps even when linear models cannot detect it.

3.4 Limitation of project outcomes

Some limitations faced by this approach are

1. The accuracy from directional testing resulted in an accuracy ranging from 0.43 to 0.5. This shows the difficulty in predicting the next day's movements using noise-filled financial data.
2. LSTM is affected by sensitive market movements which can be seen by the performance of the model when tested on UOB data.
3. The significance of sentiment analysis suggests the presence of nonlinear effects that are difficult to interpret.
4. The use of OHLCV data limits the model as it is unable to capture the intraday shifts.
5. One day prediction can become unreliable during periods of sudden market volatility.

4. Conclusion and Future improvements

This report looks at Singapore's banking sector by combining technical indicators and sentiment score. Our findings show that there are LSTM models and sentiment data can predict market trends better than random guessing. This report builds a foundation for future market trends prediction techniques. Future improvements include Introducing LSTM/GRU deep learning, exploring ML models such as XGBoost and Random Forest, integrating more macroeconomic variables (GDP, interest rates, regional indices).

5. References

Wang, Sangyuan et al. "VIX Constant Maturity Futures Trading Strategy: A Walk-Forward Machine Learning Study." *PloS one* 19.4 (2024): e0302289. Web.

Ciganovic, M., & D'Amario, F. (2024). Forecasting cryptocurrencies log-returns: a LASSO-VAR and sentiment approach. *Applied Economics*, 56(58), 8112–8138.

<https://doi.org/10.1080/00036846.2023.2289930>

The Straits Times. (2025, October 31). UOB Q3 profit tumbles 72% to \$443 million as bank raises pre-emptive provisions. <https://www.straitstimes.com/business/banking/uob-q3-profit-tumbles-72-to-443-million-as-bank-raises-pre-emptive-provisions>

6. Appendix

Fig:1 UOB

```
Call:
lm(formula = Log_Ret ~ ., data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-1.029e-03 -4.287e-05 -6.640e-06  4.469e-05  6.201e-04

Coefficients: (3 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.719e-03  2.350e-03   2.008  0.0454 *
Open         5.591e-05  6.099e-05   0.917  0.3600
High        2.370e-03  1.298e-03   1.825  0.0689 .
Low        -1.709e-03  8.836e-04  -1.935  0.0539 .
Close       2.604e-05  1.320e-04   0.197  0.8438
Volume     -4.083e-11  6.325e-12  -6.456 3.74e-10 ***
Dividends   7.494e-05  7.681e-05   0.976  0.3299
Stock.Splits NA         NA         NA      NA
MACD        1.511e-03  8.466e-04   1.785  0.0751 .
MACD_Hist   4.658e-03  2.411e-03   1.932  0.0542 .
EMA         1.077e-03  8.340e-04   1.291  0.1976
SMA         2.590e-04  1.444e-04   1.794  0.0737 .
CCI         4.108e-07  2.366e-07   1.736  0.0834 .
Aroon_Up    -1.413e-07  3.447e-07  -0.410  0.6822
Aroon_Down  -2.419e-07  3.662e-07  -0.661  0.5092
RSI         -5.023e-06  3.096e-06  -1.622  0.1057
SO          1.335e-07  7.971e-07   0.167  0.8671
MOM        -9.778e-05  2.126e-04  -0.460  0.6458
ROC         1.002e+00  6.389e-03 156.830 < 2e-16 ***
BB_High     1.338e-04  5.964e-05   2.243  0.0255 *
BB_Low      1.076e-04  5.390e-05   1.996  0.0467 *
NATR        -8.494e-05  1.258e-04  -0.675  0.5000
OBV        -1.632e-12  9.206e-13  -1.773  0.0771 .
CMF        -1.058e-05  5.680e-05  -0.186  0.8523
MFI         8.653e-08  9.570e-07   0.090  0.9280
PVT        -1.239e-10  7.207e-11  -1.719  0.0866 .
VWAP       -2.404e-04  1.396e-04  -1.721  0.0861 .
UpperBand_AB -9.497e-04  4.151e-04  -2.288  0.0228 *
MiddleBand_AB NA         NA         NA      NA
LowerBand_AB NA         NA         NA      NA
AMA        -6.587e-05  6.400e-05  -1.029  0.3041
tenkan_sen  -2.918e-06  4.511e-05  -0.065  0.9485
TEMA       -1.247e-03  8.266e-04  -1.509  0.1322
Sentiment_Score -1.967e-06  3.609e-06  -0.545  0.5862
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.000122 on 337 degrees of freedom
Multiple R-squared:  0.9999,    Adjusted R-squared:  0.9999
F-statistic: 1.01e+05 on 30 and 337 DF,  p-value: < 2.2e-16
```

Fig 2: DBS

Call:

```
lm(formula = Log_Ret ~ ., data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.886e-04	-3.565e-05	6.200e-07	3.525e-05	4.037e-04

Coefficients: (2 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.316e-03	1.568e-03	0.839	0.401957
Open	-7.268e-05	3.242e-05	-2.242	0.025764 *
High	2.606e-02	1.591e-03	16.374	< 2e-16 ***
Low	-1.733e-02	1.078e-03	-16.077	< 2e-16 ***
Close	-4.047e-04	7.242e-05	-5.587	5.54e-08 ***
Volume	-1.383e-11	2.633e-12	-5.252	3.02e-07 ***
Dividends	2.584e-05	5.653e-05	0.457	0.648001
Stock.Splits	1.681e-04	8.207e-05	2.049	0.041451 *
MACD	1.982e-03	4.642e-04	4.269	2.71e-05 ***
MACD_Hist	6.373e-03	1.325e-03	4.810	2.49e-06 ***
EMA	1.752e-03	4.620e-04	3.792	0.000184 ***
SMA	2.923e-04	7.341e-05	3.982	8.76e-05 ***
CCI	2.712e-07	1.782e-07	1.522	0.129277
Aroon_Up	-2.657e-07	3.146e-07	-0.844	0.399188
Aroon_Down	-2.946e-07	2.652e-07	-1.111	0.267556
RSI	1.837e-06	2.256e-06	0.814	0.416282
SO	3.663e-08	6.091e-07	0.060	0.952090
MOM	5.136e-04	9.130e-05	5.626	4.55e-08 ***
ROC	9.856e-01	3.267e-03	301.647	< 2e-16 ***
BB_High	8.238e-05	3.907e-05	2.108	0.035897 *
BB_Low	6.279e-05	3.513e-05	1.788	0.074937 .
NATR	-1.378e-04	1.008e-04	-1.368	0.172484
OBV	5.346e-13	5.269e-13	1.015	0.311117
CMF	-7.446e-06	4.031e-05	-0.185	0.853593
MFI	-1.086e-06	6.900e-07	-1.574	0.116694
PVT	-3.089e-11	2.598e-11	-1.189	0.235496
VWAP	-2.117e-04	1.588e-04	-1.334	0.183380
UpperBand_AB	-8.664e-03	5.195e-04	-16.678	< 2e-16 ***
MiddleBand_AB	NA	NA	NA	NA
LowerBand_AB	NA	NA	NA	NA
AMA	2.517e-05	2.514e-05	1.001	0.317641
tenkan_sen	6.967e-06	3.037e-05	0.229	0.818711
TEMA	-1.797e-03	4.543e-04	-3.955	9.74e-05 ***
Sentiment_Score	2.224e-06	1.628e-06	1.366	0.173025

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.114e-05 on 275 degrees of freedom

Multiple R-squared: 1, Adjusted R-squared: 1

F-statistic: 2.107e+05 on 31 and 275 DF, p-value: < 2.2e-16

Fig 3: OCBC

```
Call:
lm(formula = Log_Ret ~ ., data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-6.038e-04 -3.064e-05  3.550e-06  3.141e-05  7.194e-04

Coefficients: (3 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.031e-03  2.997e-03   0.344  0.731003
Open        -4.636e-04  1.052e-04  -4.408  1.41e-05 ***
High         2.014e-01  1.172e-02  17.185 < 2e-16 ***
Low         -1.341e-01  7.873e-03 -17.027 < 2e-16 ***
Close       -9.460e-04  2.178e-04  -4.343  1.88e-05 ***
Volume      -3.427e-12  2.596e-12  -1.320  0.187770
Dividends    2.453e-04  1.049e-04   2.339  0.019952 *
Stock.Splits      NA         NA      NA      NA
MACD          3.506e-03  1.428e-03   2.456  0.014579 *
MACD_Hist     1.184e-02  4.028e-03   2.941  0.003504 **
EMA           2.323e-03  1.422e-03   1.633  0.103370
SMA           4.861e-04  2.312e-04   2.103  0.036228 *
CCI           4.615e-08  1.530e-07   0.302  0.763070
Aroon_Up      -3.654e-07  2.610e-07  -1.400  0.162454
Aroon_Down    -2.081e-07  2.573e-07  -0.809  0.419261
RSI           2.052e-06  2.722e-06   0.754  0.451500
SO            -3.818e-07  6.018e-07  -0.634  0.526245
MOM           1.252e-03  3.438e-04   3.642  0.000314 ***
ROC           9.835e-01  4.771e-03  206.128 < 2e-16 ***
BB_High       3.292e-04  1.072e-04   3.072  0.002305 **
BB_Low        2.147e-04  9.484e-05   2.264  0.024235 *
NATR          -6.164e-04  2.721e-04  -2.265  0.024152 *
OBV           3.067e-13  3.065e-13   1.001  0.317758
CMF           2.533e-05  4.087e-05   0.620  0.535848
MFI           -6.058e-07  6.638e-07  -0.913  0.362097
PVT           -4.445e-11  5.199e-11  -0.855  0.393117
VWAP          -3.635e-04  4.621e-04  -0.787  0.432087
UpperBand_AB  -6.654e-02  3.856e-03 -17.257 < 2e-16 ***
MiddleBand_AB      NA         NA      NA      NA
LowerBand_AB      NA         NA      NA      NA
AMA           2.347e-04  1.037e-04   2.263  0.024313 *
tenkan_sen     2.952e-04  8.868e-05   3.329  0.000971 ***
TEMA          -3.208e-03  1.392e-03  -2.304  0.021841 *
Sentiment_Score -1.757e-06  2.152e-06  -0.817  0.414799
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.806e-05 on 330 degrees of freedom
Multiple R-squared:  0.9999,    Adjusted R-squared:  0.9999
F-statistic: 1.77e+05 on 30 and 330 DF,  p-value: < 2.2e-16
```

Fig 4: Directional accuracy

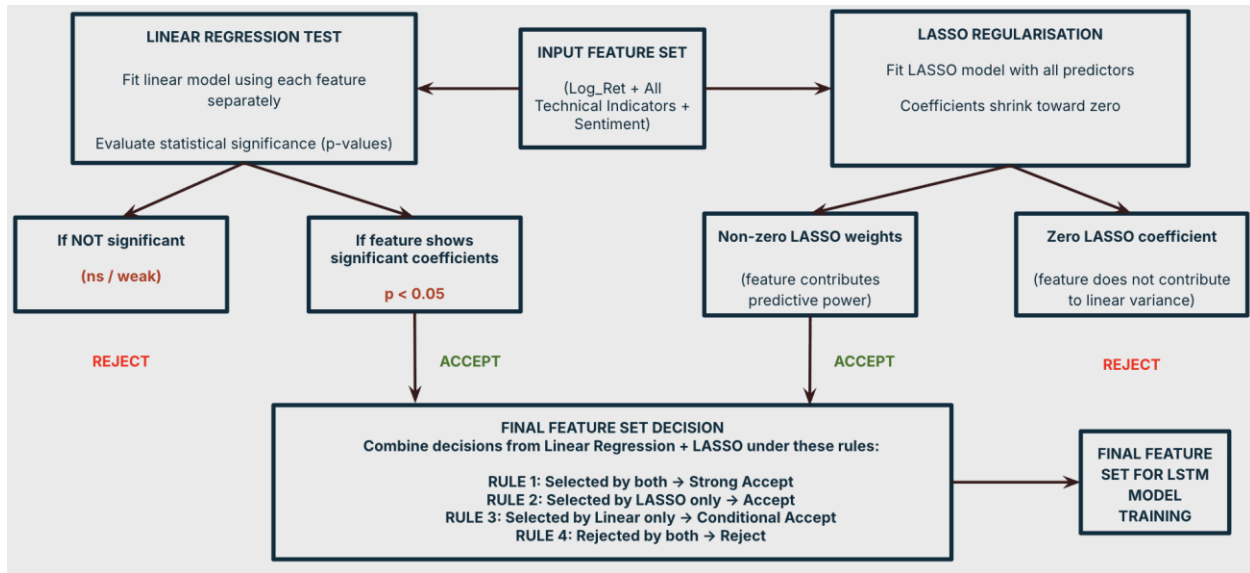


Fig 5: LSTM Model architecture

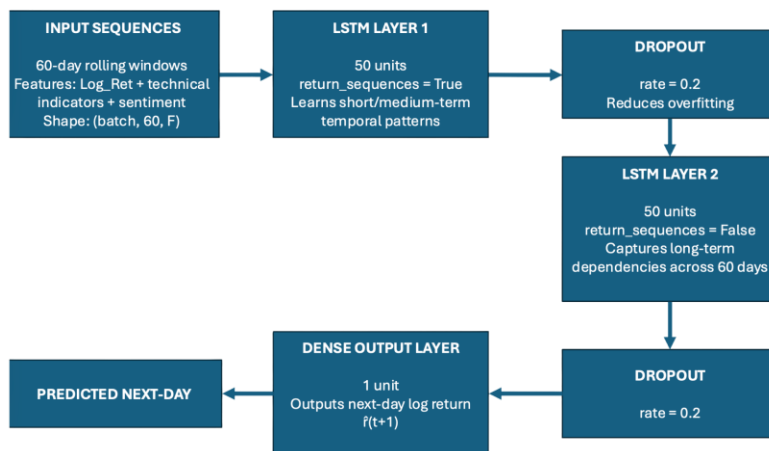


Table 1 : Linear regression significance table

Indicator	Type	UOB	DBS	OCBC
Logarithmic Returns	Return Measure	—	—	—
Moving Average Convergence Divergence	Trend	.	***	*
On-Balance Volume	Volume	(ns)	(ns)	**
Triple Exponential Moving Average	Trend (Custom)	(ns)	***	***
Ichimoku Tenkan-sen	Trend (Custom)	(ns)	(ns)	***
Exponential Moving Average	Trend	.	***	***
Simple Moving Average	Trend	(ns)	***	**
Momentum Indicator	Momentum	(ns)	***	***
MACD Histogram	Trend	.	***	**
Rate of Change	Momentum	***	***	**
Bollinger Band — Upper	Volatility	***	**	***
Aggregate Sentiment (NLP)	Custom	.	.	.

Table 2 : LASSO

Bank	Features Selected (non-zero)
O39.SI (OCBC)	MOM, ROC
D05.SI (DBS)	MOM, ROC
U11.SI (UOB)	MOM, ROC